

DELTA FORCE®

BLACK HAWK DOWN

AIN Editor Manual

Community tool v1.0

The AIN Editor is a tool designed to create navigable graphs for the AI in the game. This manual will cover everything that needs to be known about the tool, its options, how to create those navigation graphs and successfully export the files that can be used for your mission.

While most of the features present might not be used in their entirety, it's still a good idea to read everything this document has to offer so you can understand properly what the tool can and can't do.

DISCLAIMER: This tool is not affiliated with NovaLogic, its former employees, or any current owner of the Delta Force intellectual property. This is a fan-made tool designed to create files in the .ain format. No official game assets are included with the tool, and its features were developed primarily through binary analysis. A copy of Delta Force: Black Hawk Down/Team Sabre is required to use the editor fully.

Table of contents

Section 1. Feature Presentation.....	3
Section 2. Node Presentation	5
Section 3. Node Properties.....	6
Section 4. Generator and its features	7
Section 5. Graph Cleanup	9
Section 6. Takedown	10
Section 7. 3D Wireframe View	12
Section 8. Saving & Exporting AIN	18
Appendix A: Customizations	19
Appendix B: Support & Limitations.....	21
Appendix C: Key Commands	22
Appendix D. Map Samples	25

Section 1. Feature Presentation

The tool requires to read the game folder in order to load its models and terrains. The first time you open the program, please select the desired location where the game files reside. The folder can always be changed in the **‘File’** menu.

When you first open the editor, a variety of options and toggles are available. Let’s walk through each one of them:



1. The top bar – it shows the main toggles.

- **Generate** – used to generate nodes
- **Graph Cleanup** – used to delete residual nodes left by the generator
- **Export AIN** – exports the navigation graph field created as an .ain file
- **3D View** – opens the 3D wireframe viewer
- **E** – Edit toggle. Allows the option to move, delete and edit a node or a group of nodes properties
- **Pen toggle** – Draw nodes on canvas
- **4 color square** – Zone Color. Allows the option to better visualize a group of nodes by coloring them. The applied color can be used for the takedown system, however it’s not mandatory
- **AZ – Area Zones.** Used to generate specific nodes in the designated area
- **Show Zone Panel** – tied with the Area Zone. Can be toggled on/off
- **Takedown System** – Allows the created nodes to be part of the takedown system
- **X,Y** – Location on canvas
- **Zoom** – see and choose the zoom distance on canvas

2. **The left panel** – the most used part of the editor. Some of the available options that can be toggled will expand the panel further down, but it can be scrolled up/down
 - choose grid distance and snap placed nodes on the grid line
 - toggle on/off the terrain and choose what view of the terrain do you want to see
 - toggle on/off the node features
 - toggle on/off the entities you want to see on canvas
 - Z filter – toggles the ability to show nodes that are present at a certain set height
 - Info – shows the information tab for a node
 - Tiler – shows the colors used for the Zone Color/Breach system
3. **Help** – shows the most useful and available key binds.
Note: for every key bind, please read the appendix.

The editor comes with certain [?] or [!] toggles that expand upon what each option does in detail.

Section 2. Node Presentation

A **node** represents the point where the AI can move to.

An **edge** is the connection between two or more nodes.

Delta Force: Black Hawk Down uses two navigation layers for its AI movement:

- the waypoint system, which is created in MED (Mission Editor) and acts as a hard rule for the AI to follow.
- the node system, which defines where NPCs are able to move. It allows them to navigate around obstacles and through interiors, as long as a valid path is available;

With these two layers, you can tell an NPC in MED that it needs to go from waypoint 1 to waypoint 2 while the navigation graph from this editor handles how it gets there. This makes missions easier to set up in MED since the AI can rely mostly on the navigation graph to move around obstacles and through interiors.

The **radius** of a node determines how close the AI needs to get for the node to be considered reached. It is also used by the AI for certain line-of-sight checks such as teammates following the player.

There are three ways to place nodes:

- manually place nodes
- draw nodes
- generate nodes

A) Manual placement

Right mouse anywhere on canvas to place a node. Make sure the **Edit [E]** toggle is on.

B) Draw

Select the [] toggle and draw with **Left Mouse** around the map. When pressing **Right Mouse** on canvas while this mode is active you can choose the spacing and radius overlap of the nodes that are about to be drawn

C) Generate nodes

Use the [**Generate**] toggle to create fast graphs that respect an entity's boundary. Read '**Section 4. Generator and its functions**' for a more detailed overview.

Section 3. Node Properties

This section outlines the **Info** menu, located in the left panel and contains the Node Inspector. The **Node Inspector** is a menu that contains multiple fields that decide what properties a node can have along with other options that can decide what values nodes receive when placed on canvas.

- **Details** – expands the ‘Node properties’ section with additional node fields (read Chapter# Takedown System)
- **?** – built in guide for the node properties
- **ID** – each node receives a unique ID
- **X,Y,Z** – the location of a node on map
- **Control** – decides how the AI navigates. Use 0 if you want the AI to move based on radius. Use 1 if you want the AI to move towards the center of a node.
- **Radius** – choose the size of a node’s radius
- **Adaptive radius** – automatically choose the radius of a node when placed based on what’s around it
- **Place on entities** – allows the nodes to be placed on entities
- **Lock Options** – lock a node’s values or height. Every node that gets manually placed afterwards receives those locked values.
- **Neighbors** – show the selected node’s neighbors.

The screenshot shows the 'Node Inspector' window in the AIN Editor. It features a tabbed interface with 'Info' and 'Tiler' tabs. The 'Info' tab is active, displaying various node properties. At the top, there are sub-tabs for 'Node Inspector', 'Details', and a help icon. The 'Node Inspector' sub-tab is selected, showing fields for ID (1), X (-71.8467), Y (7.1658), and Z (19.7489). Below these are 'Node properties' for Control (0) and Radius (36), with an 'Apply Changes' button. Further down are checkboxes for 'Adaptive radius' (checked), 'Large Radius (48+)', 'XL Radius (128+)', and 'Place on entities'. A 'Lock options' section has 'Lock values' and 'Lock Z' checkboxes, each with an 'Update' button. Below that is a 'Neighbors' section showing 4 neighbors (Node 0, 2, 3, 4) with a 'Hide' button. At the bottom are 'Remove' and 'Connect with ID' buttons, and a large red 'Delete Node' button at the very bottom.

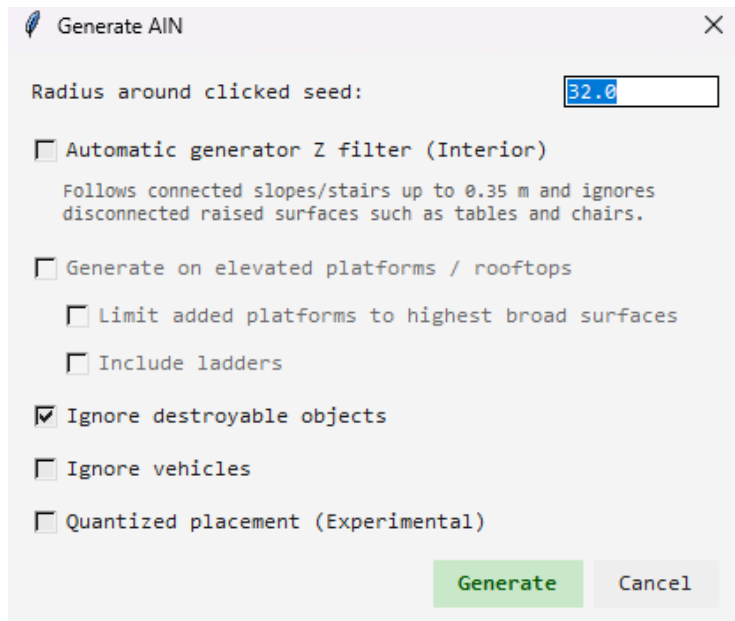
IMPORTANT! A node’s height must sit at chest level, that means terrain/building height + Z=1.0000m/1.2489m or any other value close to that range. The editor already takes care of it for the most part, but it’s something to keep in mind when deciding to work with graphs. Otherwise, the AI might fail to follow the nodes.

Section 4. Generator and its features

The generator is the most useful option for creating graphs that cover the entire playable area. When selecting the **[Generate]** toggle, a pop up will show in the middle of screen with various options.

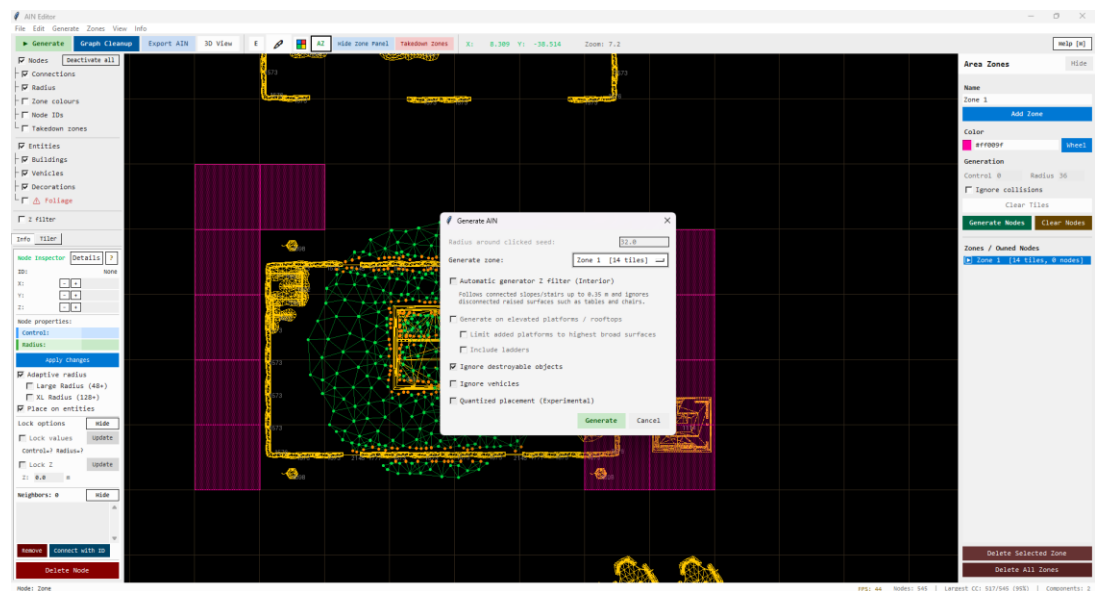
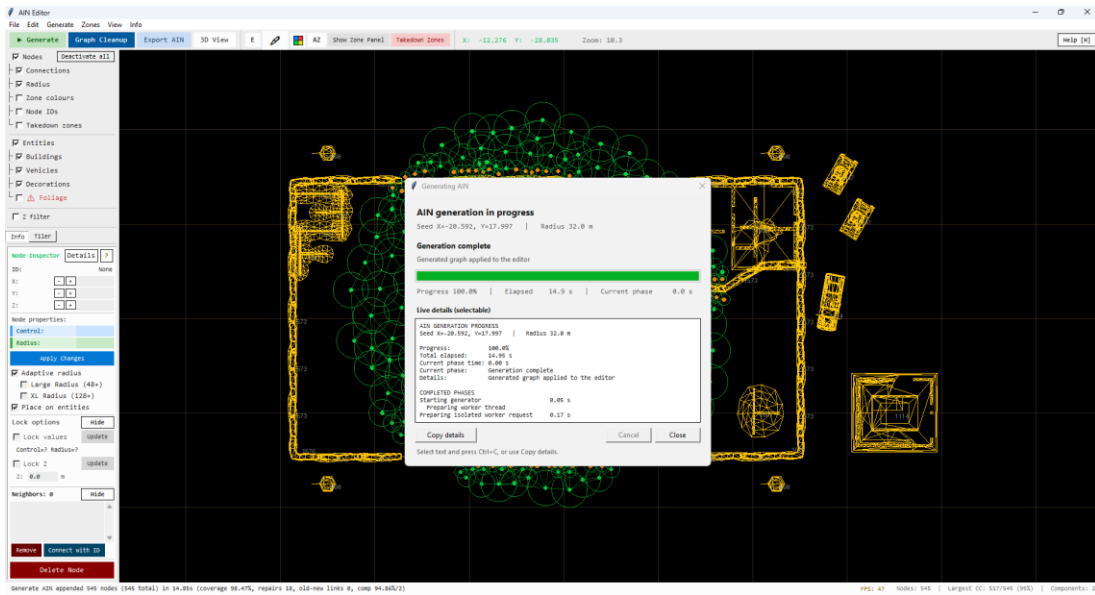
Here is an overview of what each option does:

- **Radius around clicked seed** – choose the size of the disk where nodes will get placed. Max 80m seed;
- **Automatic generator Z filter (Interior)** – allows nodes to generate inside buildings with navigable interiors;
- **Generate on elevated platforms/rooftops** – allows for nodes to generate on detected surfaces;
- **Limit added platforms to highest broad surfaces** – tied to the platforms/rooftops option. Use it if you want nodes to get placed only on rooftops and floors instead of tables and other surfaces. Certain underground segments might fail to generate with this toggle on;
- **Include ladders** – allows nodes to get placed and connect through ladders;
- **Ignore destroyable objects / Ignore vehicles** – allows nodes to ignore said object categories. Deactivate if nodes should avoid such entities;
- **Quantized placement** – experimental mode that places generated nodes using fixed height steps instead of matching the surface exactly. This is based on the placement pattern found in some of the original game navigation files.



Area Zones are grid zones defined by the user. It is the only system that bypasses the 80m seed limit. Choose if you want to generate through a seed or an area zone and wait for the process to be done. The first pass always takes slightly longer since it collects the entities present on the map.

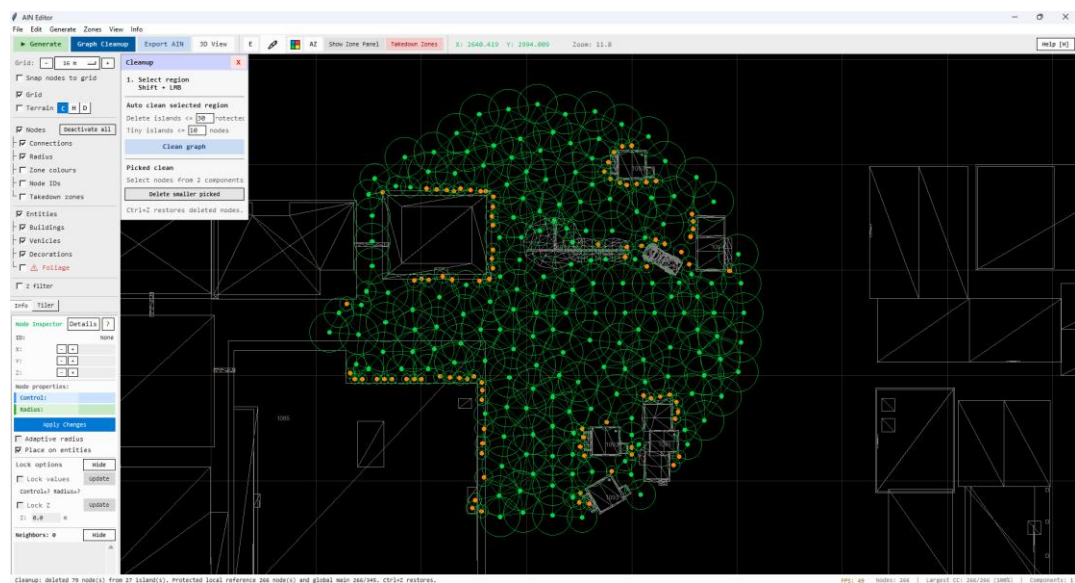
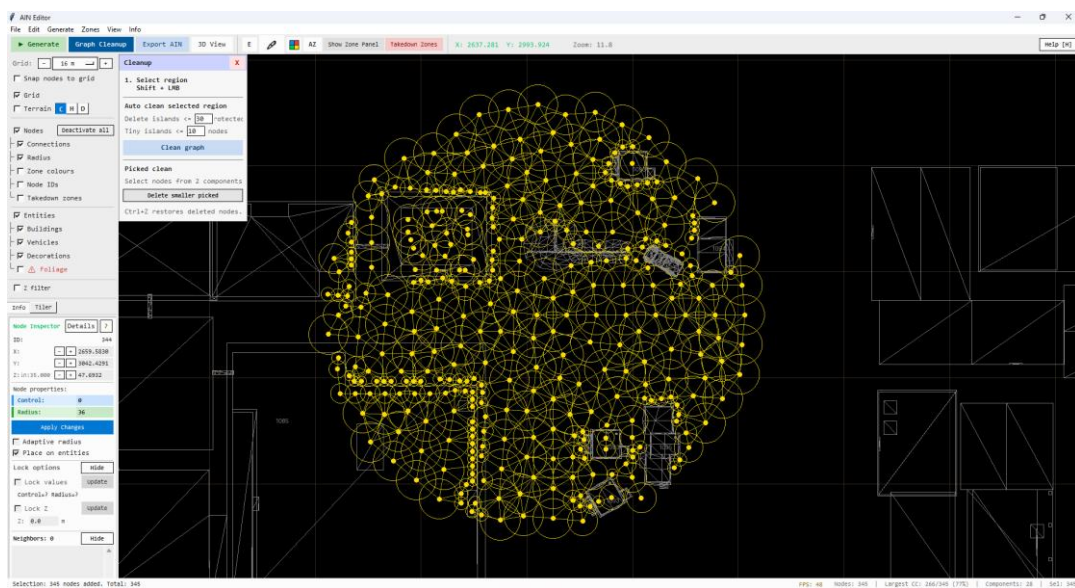
Depending on how many entities are present and if the buildings contain complex navigable interiors or not, the process could take somewhere between 30 seconds to 2 minutes. If it takes longer than 5 minutes in a small zone (Example: 32m seed), please cancel and try generating again. The larger the size, the longer the process takes.



Section 5. Graph Cleanup

The **Graph Cleanup** is a feature that deletes isolated groups of nodes. Because the generator scans for every model segment, there are a lot of situations where nodes leak through entities that have no navigable interior.

Select the graph with **SHIFT + Left Mouse Drag** and select the **[Graph Cleanup]** toggle. Choose the amount of percentage that's separated from the main graph that you want removed. Use the **'Delete smaller picked'** toggle by selecting a few nodes from the main graph and the isolated graph you want removed.



Graph Cleanup – Before vs After

Section 6. Takedown

This section covers the **Takedown** system in its entirety as it is a system that's closely tied to the node graph.

Expanding the Node Inspector through the [Details] toggle will reveal 3 new node properties:

Node properties:	
Control:	0
Radius:	32
Takedown (var1):	6
Takedown Zone:	 6
Takedown (var2):	0

- **Takedown (var1)** – defines the role of a node
 - **Takedown Zone** – defines the zone where this system can be active
 - **Takedown (var2)** – defines the zone where the system gets triggered
- These three properties will determine where and how an active takedown zone plays out.

Here are the available values that can be used for each property:

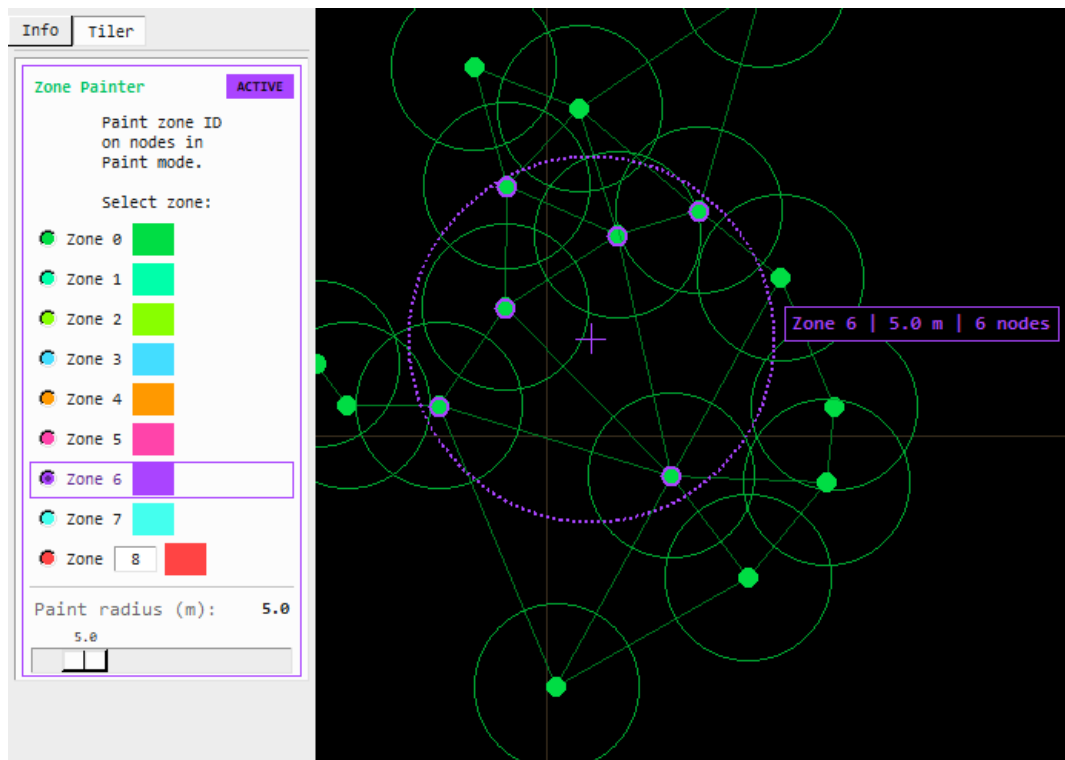
TAKEDOWN (VAR1) – use it along with the active takedown zone.

- 0 – normal (color changes based on what Takedown Zone ID is chosen) ●
- 6 - assault position. It is used to define specific locations where the NPC moves during assault ●
- 8 - grenade target. An NPC finds line of sight of this node and throws a grenade towards it. Place it relative to the entrance of a building, otherwise the flashbang animation could trigger in air or other awkward places ●
- 9 – red floating pyramid marker. Visible only during manual activation. ▲



Takedown Marker

TAKEDOWN ZONE – use any value from 1 to 255 to determine a takedown zone. Select multiple nodes or paint an area using the **Color** (■) to apply it.



Zone Color

TAKEDOWN (VAR2) – use it along takedown (var1) to determine the entrance area.

- Takedown (var1) = 2 (trigger)
- Takedown (var2) = Takedown Zone ID

IMPORTANT! Takedown Zone ID field must be 0 or any other value that does NOT match Takedown (var2). Otherwise, the trigger will fail.



Takedown Zones menu

Use the red Takedown Zones toggle situated at the top bar to assign an active takedown zone.

A zone can be triggered **Auto**, **Manual** or **Both**.

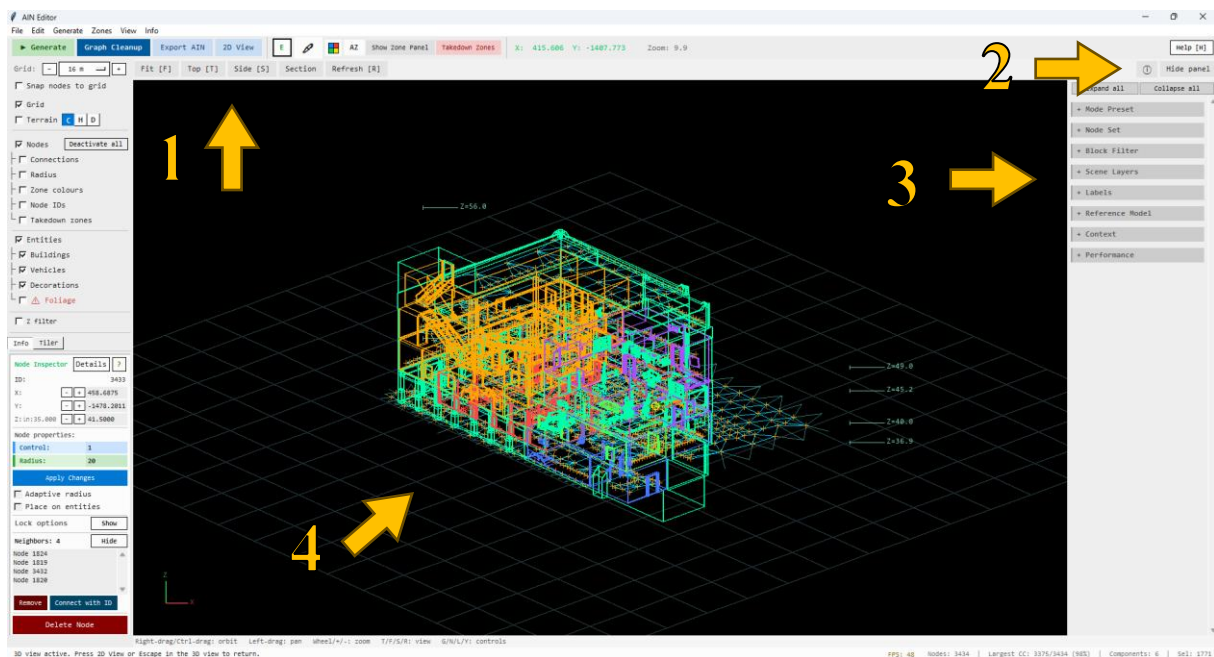
Manual is the only option where an AI can do the wall cover grenade throw animation.

Toggle on the left panel the '**Takedown zones**' to visualize only the active zones. Toggle '**Zone colors**' if no active zone is available.

Section 7. 3D Wireframe View

This section will present everything about the 3D View and the tools that are available with it.

The **3D Wireframe View** is the most complex system this editor has to offer. Unlike MED, a navigation graph could have issues, especially when it comes to anything that contains a navigable interior (oil rig platforms, garages, hotels etc.). Because of that, this system was designed with the scope of working inside such buildings.



When opening the 3D View, the editor changes to a 3D wireframe layout and reveals additional menus and controls used specifically for this view.

1. Top bar – contains toggles that modifies the camera view.
 - Fit [F] - adjust the 3D View so the visible graph stays on screen
 - Top [T] - look at the view from the top
 - Side [S] – face the view from any side
 - Section – filter the Z and X,Y levels of a building. Used extensively for complex buildings where a room is better to look at without its surroundings. Use the attached squares of each filter to move up/down/left/right
 - Refresh [R] – refresh the 3D View.



Section active

2. Hide right panel, hide guide for each 3D View menu
3. Right Panel – contains different visual and workflow methods
 - Mode preset – choose which visual present should be active
 - Node set – lock the selected group of nodes to the 3D View
 - Block filter – choose how a model gets rendered and what segments should be shown/hidden
 - Scene layers – filter what gets rendered in this view
 - Labels – tools used to identify a node ID, the height of floors or how thick should the edges of a node be
 - Reference model – places next to a node a character model that can be rotated around said node. Used to identify if an NPC can fit in a room/doorway/corridor
 - Context – how many entities get rendered
 - Performance – controls how much geometry remains visible while moving in the 3D View and limits the maximum number of lines that can be drawn.
4. Wireframe View – shows the selected nodes and visible entities

To better understand what this view can and can't do, each menu in the right panel will be explained thoroughly.

Mode Presets allow for the view to change how entities are visualized. This option contains 3 presets:

- **Normal** – makes the entities show their structure + faint shades of wires representing the walls of a building
- **Interior** – makes entities show only their structure/skeleton
- **Full view** – shows everything around the node in a large radius. It can cause massive performance issues

Each preset can be modified and saved depending on preferences. They can always be reverted to their initial options.

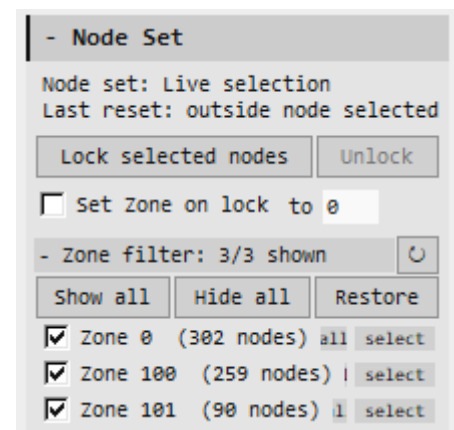


Node set is one the most important and useful options available in the 3D View. It allows for the selected nodes to get locked to the view and be edited. The following features are available while a Node Set is locked:

- select a node and edit its values. Multiple nodes can be selected with **SHIFT + Left Mouse**
- connect a selected node by using **Right Mouse** on it – **Connect from node ID** and selecting the desired node you want to connect with
- change their location through the **Node Inspector's X,Y,Z** fields
- undo/redo a change with **CTRL + Z / CTRL + Y**

What this option can **NOT** do:

- move nodes around freely with mouse (see Appendix C: Key Binds)
- create nodes



Node Set menu with active zones

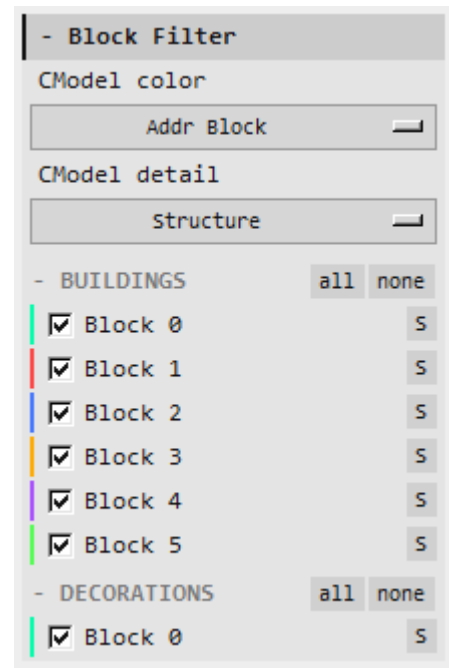
The node set allows for filtering group of nodes based on their takedown zone ID. It can be useful if a room or floor is designed to have a zone ID that can be easier to see/hide in this 3D wireframe view.

The **Block filter** allows for entities to change how they are rendered.

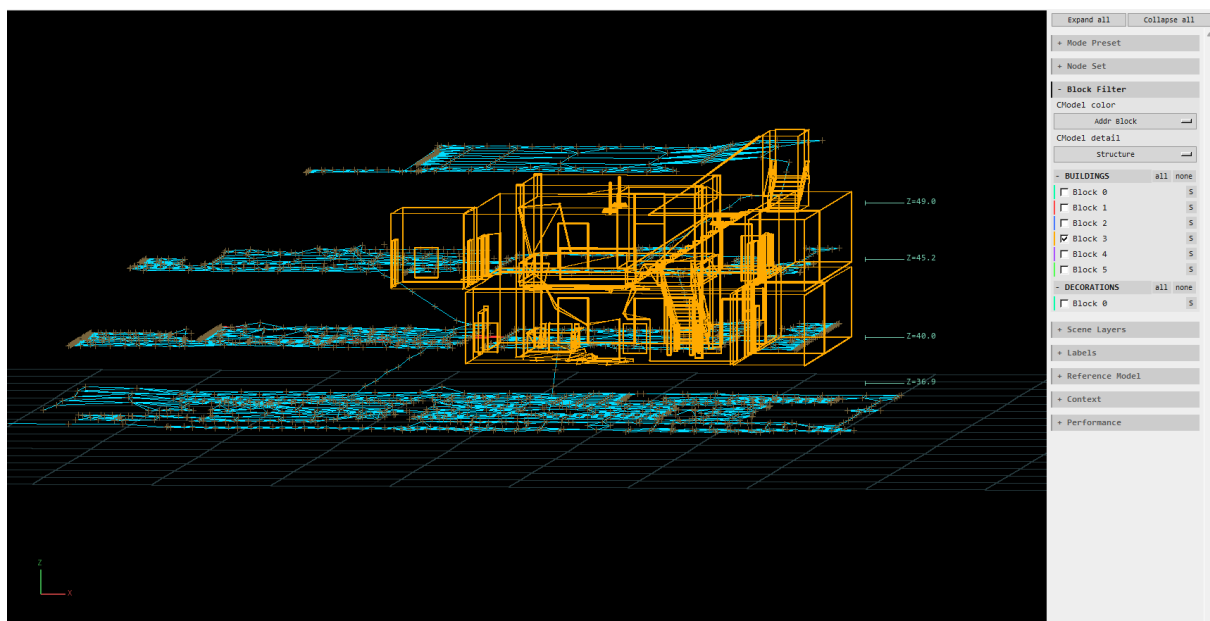
CModel color allows for the wireframes to be rendered either through its source model or its block models. Default is Addr Block.

CModel detail allows for the structure of an entity to hide or render the wireframes.

- **Raw wire** allows entities to render every single wireframe.
- **Structure** allows entities to render only the skeleton of the building. This is the only option that will **not** render the **CModel color** – **Source**
- **Structure + ghost** allows entities to render the skeleton + wireframes that defines the walls



This menu has the ability to filter what entity block's (Addr Block) get rendered. It can be helpful when a building becomes a soup of colors and using **Section** could prove to be too complicated. However, unlike Section, this option does not hide the nodes that aren't part of the selected group of nodes.



Block filter active

Scene layers is a menu that contains a mix of entity filter options and node related options.

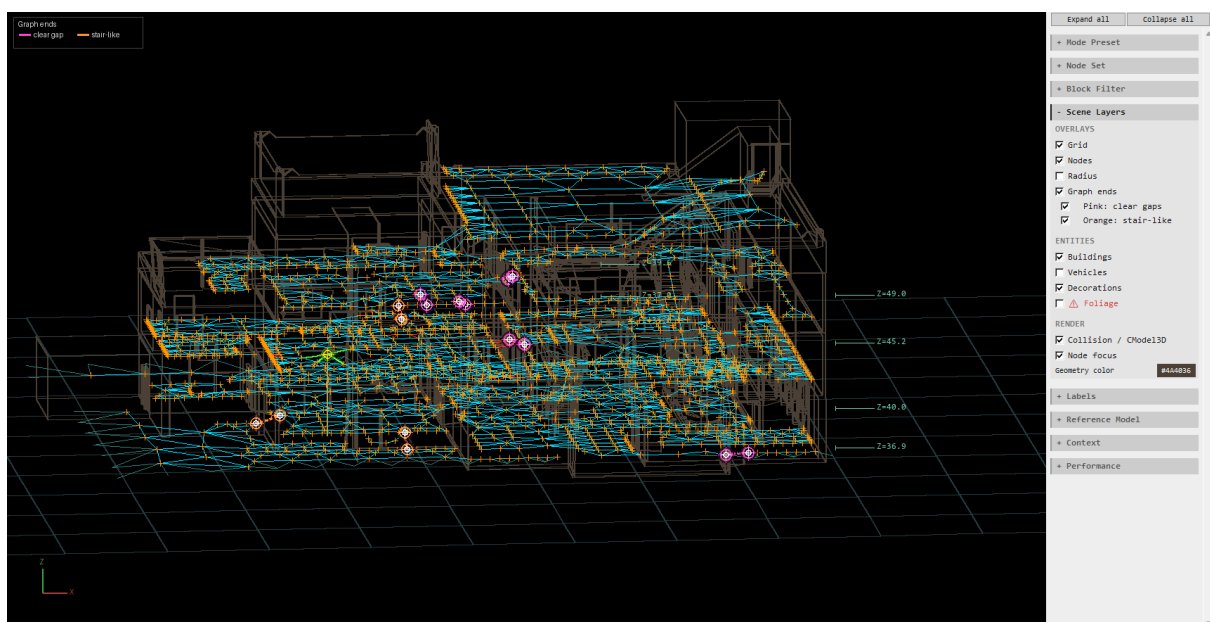
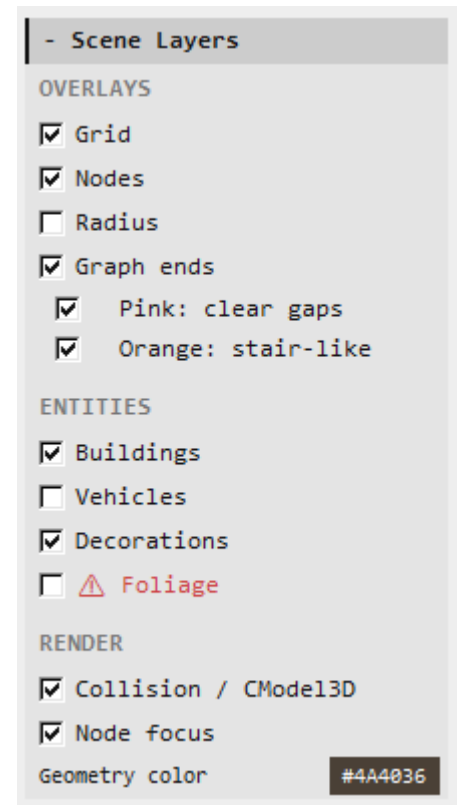
OVERLAYS contain options such as:

- **Grid** – show or hide the grid
- **Nodes** – show or hide the nodes present in this view
- **Radius** – show or hide a node's radius. Depending on how many nodes are selected, the visible radii caps at 64.
- **Graph Ends** – highlights places where two disconnected parts of the navigation graph end unusually close to each other. It is useful for finding missing connections or small gaps in the graph. The editor visualizes them in two separate ways:
 - **Pink** – clear gaps between disconnected graph ends
 - **Orange** – stair-like gaps, where height changes may indicate a missing stair connection

ENTITIES show or hide buildings, vehicles, decorations or foliage.

RENDER contains toggles related to rendering.

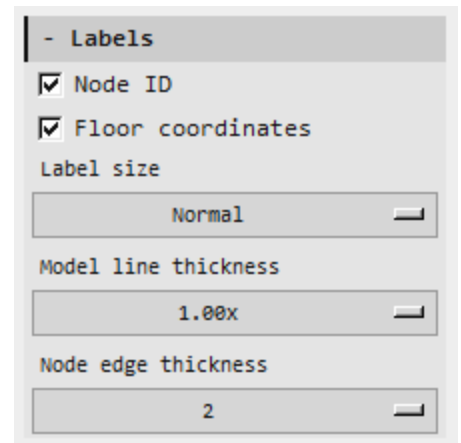
- **Collison / CModel 3D** can show or hide every entity wireframes.
- **Node focus** changes every entity into a dark brown color and renders the nodes on top of the entities. It is very useful when trying to focus on the graph and its problems rather than looking at each entity block segments.



Node focus + Graph ends active

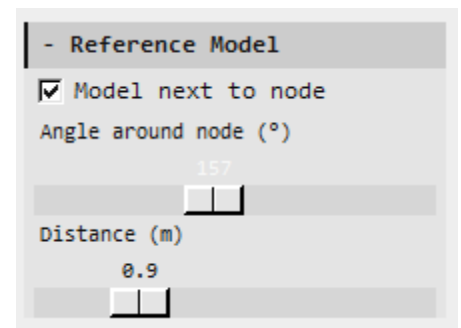
Labels contain options related to node identifiers and view helpers such as:

- **Node ID** – show or hide the unique ID of the available nodes
- **Floor coordinates** – renders light teal indicators that detects where floors/rooftops are present in a building. Use the Side [S] option at the top bar for better visuals of the heights and accuracy. It can be used to determine where a node could sit in relation with its height.
- **Label size** – change the size of the rendered labels from small, normal and large.
- **Model line thickness** – determines how thick the wireframes of an entity are to x0.50 or x.1.0
- **Node edge thickness** – determines how thick the edges of nodes are rendered from x1, x2 and x3.

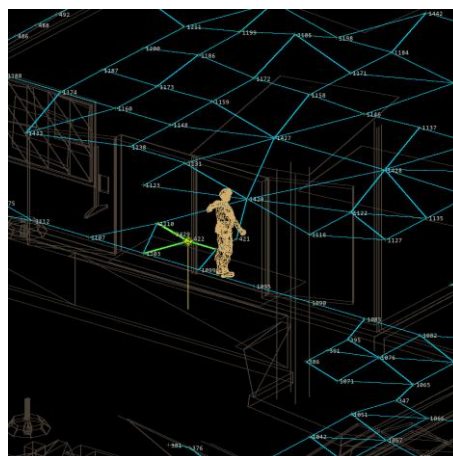
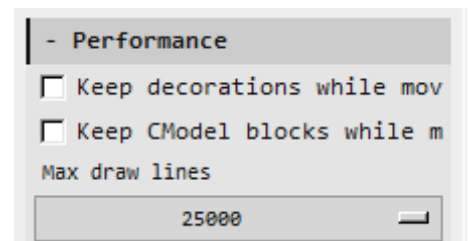


Reference model renders a 3D NPC model next to a selected node. It is useful when determining if a node is too high/low or does not fit in a certain space.

Context determines which type of entities are rendered at a time. It contains 5 presets dedicated to this menu such as **Focus**, **Local**, **Near-** and **Near+** which increase how many entities can be present and **Full** which showcases everything it can.



Performance menu contains options that affect positively or negatively the 3D wireframe view depending on what is active. Because the render can get expensive and performance issues could arise, this menu allows or denies entities to render while rotating/moving around. It also contains a limit that determines how many wireframes are allowed to be rendered with default being 15000.



Reference model

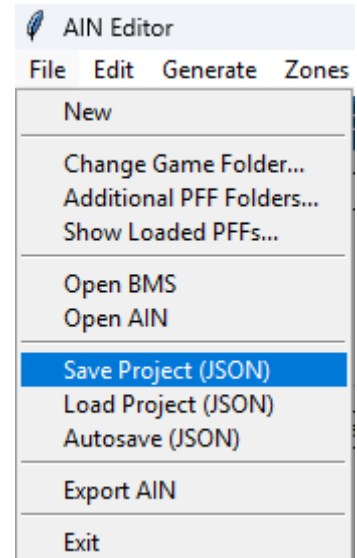
Section 8. Saving & Exporting AIN

Save the project as a .json file from **File - Save Project (JSON)** if you want to preserve generator seed history and area zones.

The **Autosave (JSON)** option allows for the saved project to autosave every 5 minutes.

Export the graph either through **File – Export AIN** or use the blue toggle at the top bar with the same name to export said file.

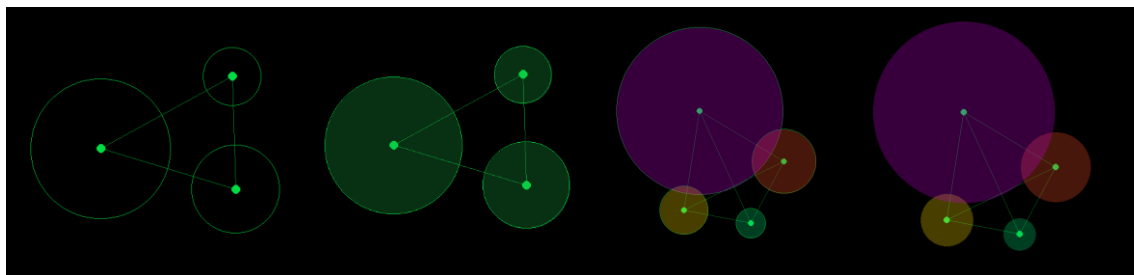
Make sure the file name matches the .bms map name.



Appendix A: Customizations

The editor allows certain visual customizations that can be changed based on personal or workflow preferences.

- UI – change from light to dark mode. This applies to menus/panels, canvas or both.
- Radius – choose how the radius gets visualized on graph.
 - Show only the outline of the radius
 - Show radius + fill (alpha overlap)
 - Show radius + different fill color based on the size of the radius
 - Keep radius when the outline is off
- Hide arrow indicator present when the left panel can be scrollable
- Show/hide water levels
- Show/hide terrain
- Trees only
- Change the color of entities in light/dark mode

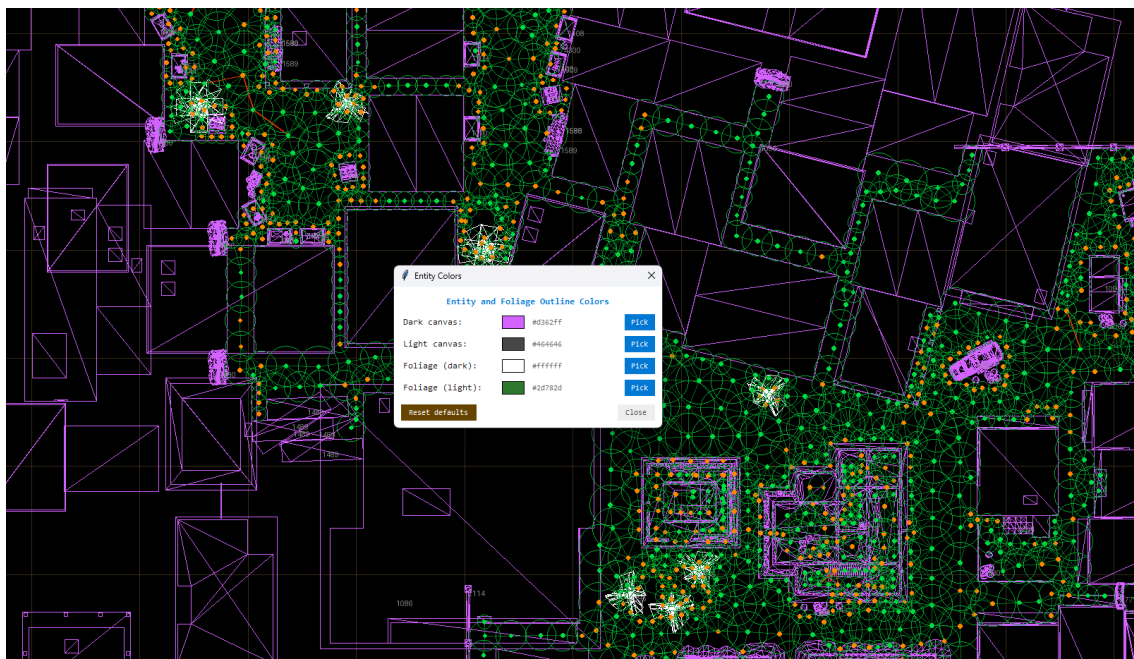


Outline only

Fill

Color by size fill

Fill without outline



Entity color change



Dark mode



Light mode

Appendix B: Limitations

1. The editor can run only on **Windows 10/11**.
2. If there are problems with entities not being rendered, please make sure the game is **set in the right folder**.
3. Mod support is available. (tested: NSO Deadline, TIO Coop)
4. Foliage heavy maps will make the editor lag/stutter in both 2D and 3D view.
5. The generator is not perfect. It's a fast method to place hundreds or thousands of nodes in a short amount of time. The editor allows the ability to remedy the mistakes left by it.
6. The **quantized** method of the generator is strictly an experimental method and might cause edge and node placement problems at certain entity transitions.
7. The editor offers a **developer mode**. It contains binary info and debug panels that were used thorough the development of it.
8. Because of the way the canvas is rendered, it is better to move around with the mouse than with the arrow keys.
9. A seed might fail to generate depending on where it was placed (happens when the seed is placed over a wall inside a building with many floors)
10. Any entity that is pitched and navigable (example: open cargos) will create edges going through the model. Use the 3d view to identify and remove them (marked with a red line). This is a known generator bug and the fix breaks other generator sub systems.

Appendix C: Key Commands

These are all the available keys in the AIN Editor:

AIN Key

Chart

Mode	Keyboard	Mouse	Effect
2D			
	E		Switch to edit mode
	D		Switch to draw mode
	C		Switch to Color/paint mode
	Z		Switch to Area Zones
Draw	R		Toggle Radius Brush
Edit	CTRL + R		Rebuild neighbors for selected nodes
Edit	U		Clear node selection
Edit	F		Fit all nodes in view
	G		Toggle grid
	H		Toggle help panel
	F1		Terrain display: Color
	F2		Terrain display: Height
	F3		Terrain display: Depth
	F8		Open Zone Layer Manager
Dev	F9		Toggle debug bar
Edit	CTRL + Z		Undo action
Edit	CTRL + Y		Redo action
Edit	CTRL + C		Copy
Edit	CTRL + V		Paste
Edit	Delete		Delete selected nodes/area zones
	Escape		Cancel current action
Edit	LEFT		Move node to the left 1 meter
Edit	RIGHT		Move node to the right 1 meter
Edit	UP		Move node up 1 meter
Edit	DOWN		Move node down 1 meter
Edit	SHIFT + LEFT		Move node to the left by 10 cm
Edit	SHIFT + RIGHT		Move node to the right by 10 cm
Edit	SHIFT + UP		Move node up by 10 cm
Edit	SHIFT + DOWN		Move node down by 10 cm
Edit	CTRL + LEFT		Alternative move left 1 meter
Edit	CTRL + RIGHT		Alternative move right 1 meter
Edit	CTRL + UP		Alternative move up 1 meter
Edit	CTRL + DOWN		Alternative move down 1 meter
	NUMPAD 9		Zoom in
	NUMPAD 3		Zoom out
	NUMPAD 8		Pan up

	NUMPAD 2		Pan down
	NUMPAD 4		Pan left
	NUMPAD 6		Pan right
	NUMPAD +		Increase grid size
	NUMPAD -		Decrease grid size
	+		Alternative increase grid size
	-		Alternative decrease grid size
Edit		L click node	Select/deselect a node
Edit		L drag node	Move a node or selected group of nodes
	A		Apply changes from Node Inspector
	U		Clear node selection
Edit		L drag canvas	Pan
		Middle drag	Pan
	ALT	L drag canvas	Pan
		Mouse wheel	Zoom around the pointer
Edit	SHIFT	L drag	Box select additional nodes
Edit	CTRL	L drag	Removes nodes from selection using a box
		R click node	Open node's context menu
Edit		R click canvas	Place node
Edit	CTRL	R drag	Connect nodes by drawing a brush stroke
Edit	SHIFT	R drag	Cut connections touched by the stroke
Draw		L drag	Place nodes
Draw		R click	Open draw spacing / radius brush options
Color		L click/drag	Paint zone IDs onto nodes
Area Zones		L click/drag	Select tiles
Area zones		R click/drag	Erase tiles
		L click generator seed	Generate nodes
		R click generator seed	Cancel the generator seed
3D			
	T		Top view
	S		Side view
	F		Fit view
	G		Toggle grid
	U		Clear node selection
	N		Toggle node overlay
	C		Toggle buildings, vehicles and decorations together
	B		Toggle buildings
	V		Toggle vehicles
	D		Toggle decorations
	R		Refresh the 3D view
	L		Toggle floor Z levels
	Y		Toggle node IDs

	[		Previous active node in the scene
	]		Next active node in the scene
	+		Zoom in
	=		Zoom in alternative
	-		Zoom out
	NUMPAD 9		Zoom in
	NUMPAD 3		Zoom out
	NUMPAD 4		Move node to the left by 1 meter
	NUMPAD 6		Move node to the right by 1 meter
	NUMPAD 8		Move node to the front by 1 meter
	NUMPAD 2		Move node to the back by 1 meter
	SHIFT + NUMPAD 4		Move node to the left by 10 cm
	SHIFT + NUMPAD 6		Move node to the right by 10 cm
	SHIFT + NUMPAD 8		Move node to the front by 10 cm
	SHIFT + NUMPAD 2		Move node to the back by 10 cm
	CTRL + Z		Undo
	CTRL + Y		Redo
		L drag	Pan
		R drag	Orbit
	CTRL	R drag	Orbit alternative
		Mouse wheel	Zoom around pointer
	SHIFT	L drag	Select one or multiple nodes with box selection
	CTRL	L drag	Remove nodes with box select
		L click node	Select node
		L click empty space	Clear live selection
		R click selected node	Open node context menu
		L click during connection	Complete the connection
	ESCAPE		Cancel connection
	ESCAPE		Exit 3D View

Appendix D. Map Samples

The editor comes with two maps designed to test how the AI moves around the graph and how the takedown system gets triggered.

- **AIN_Test.bms** – a small sized map with a few buildings and obstacles that was used through the development of this editor. Use it to test the behavior of NPCs,
- **AIN_Takedown.bms** – a small sized map designed to look like a tactical assault event and used to test the takedown system.
- **AIN_Takedown.ain** – the navigation graph of AIN_Takedown.bms. Use it to test this system by changing node placements inside the takedown zone or modifying them.

